

LESSON 11.5

IS IT PROPORTIONAL?



LESSON INTRODUCTION

MA.7.AR.4.1 Determine whether two quantities have a proportional relationship by examining a table, graph, or written description.

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

LEARNING TARGETS

- I can determine if two quantities have a proportional relationship.
- I can determine the constant of proportionality of a proportional relationship.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

WORKING TOWARD

MA.8.AR.3.1 Determine if a linear relationship is also a proportional relationship.

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.

ESSENTIAL QUESTIONS

- How do you determine if a relationship is proportional from a graph, table, or written description?
- How can you use a graph, table, or written description of a proportional relationship to determine the constant of proportionality?

LESSON NARRATIVE

This lesson builds on the previous lesson, where students used graphs and tables to determine if relationships are proportional or not. After practicing determining whether relationships are proportional from graphs and tables, students will be introduced to how to determine whether a relationship is proportional from a written description. Students will apply learning from prior lessons to find and interpret the constant of proportionality for relationships that are proportional.

COMPONENT SUMMARY

11.5.1 WARM-UP (~5 MIN)

Describe a visual pattern

11.5.3 GUIDED INSTRUCTION (10 MIN)

Use written descriptions to determine if a relationship is proportional

11.5.5 WRAP-UP (~5 MIN)

Determine if a table shows a proportional relationship

11.5.2 GUIDED PRACTICE (10-15 MIN)

Use tables and graphs to determine if a relationship is proportional

11.5.4 TRY IT (10-15 MIN)

Determine if relationships are proportional

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Constant of proportionality

MATERIALS

- (Optional) Graph paper

ADDITIONAL PREPARATION

- Consider allowing students to use discs as they explore the pattern in the 11.5.1 Warm-Up.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may fail to check the ratio between all quantities in a table of values and erroneously state a relationship is proportional because some of the ratios are equivalent.
- Students may struggle to determine pairs of values from a written description.

THINGS TO CONSIDER

- Have students identify the ratios of each pair of values in a table.
- Encourage students to create a table for the pairs of values in a written description.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

The 11.5.3 Guided Instruction invites students to extrapolate information from a written description to solve a mathematical problem. Demonstrate your thinking out loud about the process of determining the pairs of values and finding the ratio of each. Then encourage students to reflect on the process, using your example as a bridge to their own thinking in future components.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

The 11.5.4 Try It provides several opportunities for students to explore relationships in real-world contexts to determine whether they are proportional. Support students as they use models and methods to connect mathematical concepts to everyday experiences. Encourage students to think about how the concepts explored in the lesson can be applied to other real-life experiences.

DIFFERENTIATE INSTRUCTION

The graphs and tables utilized in the 11.5.2 Guided Practice and 11.5.4 Try It can assist students who benefit from a visual representation of a concept. Encourage students to create one of these visual representations when given only a written description if that helps them determine the type of relationship. This also provides feedback about whether they are understanding the contexts they are reading. Auditory learners would benefit from the teacher thinking aloud through the 11.5.3 Guided Instruction while modeling the thinking used to determine whether a relationship is proportional from a written description.

11.5.1 WARM-UP: VISUAL PATTERN

DIFFERENTIATE INSTRUCTION

Consider allowing students to use discs or other objects to model the patterns. Being able to move and manipulate objects may help students who benefit from kinesthetic learning opportunities.

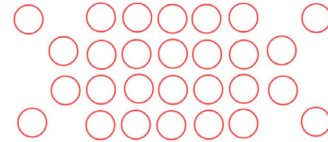


11.5.1 Warm-Up: Visual Pattern

- The first 3 steps in a pattern created using Froot Loops are shown.



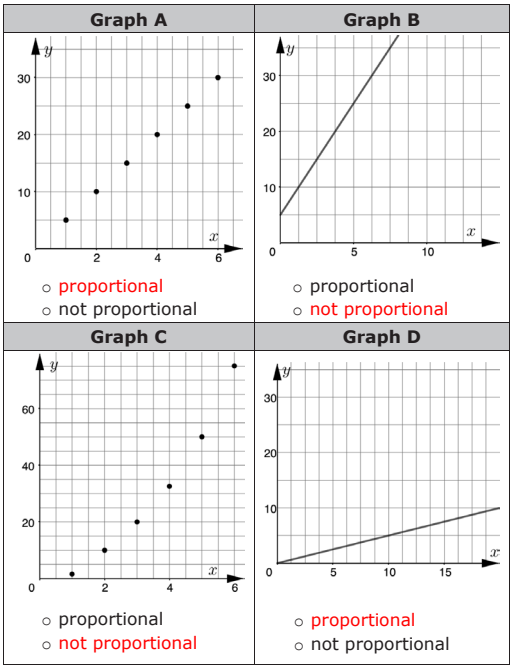
- Describe how the pattern is changing in each step.
A vertical line of 4 fruit loops is inserted in the middle of the fruit loops in each step.
- Draw what you think the 6th step in this pattern will look like using small circles to represent each Froot Loop.
Answers will vary.





11.5.2 Guided Practice: Determine if a Relationship is Proportional from Tables and Graphs

1. Four graphs are shown in the chart. Indicate whether each graph represents a proportional relationship.



11.5.2 GUIDED PRACTICE: DETERMINE IF A RELATIONSHIP IS PROPORTIONAL FROM TABLES AND GRAPHS

TEACHER MOVES

Encourage students to explain how they can tell if a graph shows a proportional relationship. Consider having students work with a partner to talk through each graph and the criteria to determine if it shows a proportional relationship.

For a proportional relationship, the constant of proportionality can be determined from a graph in two ways.

- Use the point $(1, k)$, where k is the constant of proportionality.
 - Find the value of the ratio $\frac{y}{x}$ using any point on the graph.
2. Two of the graphs in the previous question represent proportional relationships. Complete the statements to state the value of the ratio between the two quantities in each of those two graphs.
- a. Graph **A** represents a proportional relationship where the constant of proportionality is **5**.
 - b. Graph **D** represents a proportional relationship where the constant of proportionality is $\frac{1}{2}$.
3. Dylan and Maya were running laps around the football field at Hernando High School. The coach recorded their times at the end of laps 2, 4, 6, and 8.
- a. Complete the table to find the ratio of minutes to laps for Dylan throughout his run.

Dylan's Run		
Distance (laps)	Time (minutes)	Minutes per Lap
2	4	2
4	10	2.5
6	15	2.5
8	23	2.875

TEACHER MOVES

Encourage students to explain how they can calculate the constant of proportionality solely from the graph. Consider having students work with a partner to talk how they would teach an absent student how they arrived at their answers for question 1.

11.5.2 GUIDED PRACTICE: DETERMINE IF A RELATIONSHIP IS PROPORTIONAL FROM TABLES AND GRAPHS (CONTINUED)

TEACHER MOVES

Help students understand the relationship of a ratio to a constant of proportionality. Have students compare the minutes per lap for Dylan and Maya. Guide students to understand that every pair will have a ratio, but the ratios may not all be equivalent. Discuss why this may make sense in a real-world context. If the ratio is the same for every data pair in the table, then that ratio is the constant of proportionality for the entire table.

- b. Complete the table to find the ratio of minutes to laps for Maya throughout her run.

Maya's Run		
Distance (laps)	Time (minutes)	Minutes per Lap
2	5	2.5
4	10	2.5
6	15	2.5
8	20	2.5

- c. Complete the statement.

The table of values representing
☐ Dylan's
☒ Maya's
 run is a proportional relationship. The constant of proportionality is 2.5.



When determining whether two quantities have a proportional relationship from a table, **check every pair of values** in the table to determine if the ratios between them are all equivalent.



11.5.3 Guided Instruction: Determining if a Relationship is Proportional from a Written Description

To determine if a relationship is proportional from a written description, identify pairs of values that are part of the relationship based on the information given. It is helpful to consider at least 3 or 4 pairs of values, if possible.

1. The cooking time needed when using a pressure cooker changes depending on the elevation above sea level. Cooking time should be increased by 5 percent for every 1,000 feet (ft) after 2,000 ft above sea level.
 - a. By what percentage should the cooking time be increased when cooking at an elevation of 1,000 ft?
The cooking time does not need to be increased because it is not over 2,000 ft above sea level. So, you would increase by 0%.
 - b. By what percentage should the cooking time be increased when cooking at an elevation of 2,000 ft?
At 2,000 the cooking time still does not change because it needs to be over 2,000 ft for it to increase, therefore; it would still be 0%.
 - c. By what percentage should the cooking time be increased when cooking at an elevation of 3,000 ft?
The cooking time would need to be increased by 5%.
 - d. Write the ratio of percentage increase to elevation in cooking time from Parts A-C.

$$a = \frac{0}{1000}$$

$$b = \frac{0}{2000}$$

$$c = \frac{5}{3000} \text{ or } \frac{1}{600}$$

- e. Find the value of each ratio using a calculator.

$$a = \frac{0}{1000} = 0$$

$$b = \frac{0}{2000} = 0$$

$$c = \frac{5}{3000} \text{ or } \frac{1}{600} \approx .0016$$
- f. Explain whether the relationship between the elevation and cooking time is proportional.
The relationship between the elevation and the cooking time is not proportional because the ratios are not all equivalent.
2. The actual distance between Palm Beach, FL and Melbourne, FL is 113 miles (mi). The actual distance between Melbourne, FL and Kissimmee, FL is 55 mi. On a map of Florida, the distance between Palm Beach and Melbourne is 28.25 centimeters (cm) and the distance between Melbourne and Kissimmee is 13.75 cm.
 - a. Which two quantities are being compared?
The actual distance between Palm Beach and Melbourne in feet and the scale distance on the map in centimeters.
 - b. Identify pairs of values of the two quantities being compared.
(28.25 cm, 113 mi) and (13.75 cm, 55 mi)
 - c. Determine whether the relationship between the two quantities is proportional. If it is proportional, state the constant of proportionality. If it is not proportional, explain why not.
Answers will vary.
It is a proportional relationship because the ratios, $\frac{y}{x}$, are equivalent. I know this because when simplifying both ratios are equivalent to 4 or $\frac{1}{4}$.

11.5.3 GUIDED INSTRUCTION: DETERMINING IF A RELATIONSHIP IS PROPORTIONAL FROM A WRITTEN DESCRIPTION

TEACHER MOVES

Compare and Connect

Consider introducing the activity by utilizing a Compare and Connect routine. Use a Think Aloud to demonstrate the process of determining the pairs of values and finding the ratio of each. Provide students the opportunity to reflect on the process you used.

ENRICHMENT

After modeling how to use written descriptions to determine if relationships are proportional, consider deepening students' understanding by asking questions like:

- Why do you think that the cooking time in a pressure cooker must change with elevation? How else could you use proportional relationships in cooking or baking?
- Why do maps use proportional relationships? What would a map look like if it did not use a constant of proportionality? How would that affect a person's ability to use a map?

DIFFERENTIATE INSTRUCTION

Contexts from earlier units where students worked with ratios and proportional reasoning are deliberately spiraled into this unit to provide a clearer connection to prior content as students are being introduced to new content. Take advantage of the familiar contexts to help make connections for students who may be struggling.

11.5.4 TRY IT: DETERMINING IF A RELATIONSHIP IS PROPORTIONAL

ENRICHMENT

If you notice students are struggling to determine if the relationship is proportional, provide further scaffolding through questioning.

- What is the ratio of 20 ft to 6 miles? What is the ratio of 45 feet to 9 miles? 70 feet to 11 miles?
- If you are trying to determine if a relationship is proportional, what needs to be true of the ratios of the quantities compared?



11.5.4 Try It: Determining if a Relationship is Proportional

- The table shows the distance from which a lighthouse is visible to a boat at sea.

- Determine whether the table represents a proportional relationship.

Height of a Lighthouse (feet)	Distance from which it can be Seen (miles)
20	6
45	9
70	11
95	13
150	16

- Complete the statements.

The table ☐ does ☒ does not represent a proportional relationship.

- ☒ It does not have a constant of proportionality.
☐ The constant of proportionality is _____.

DIFFERENTIATE INSTRUCTION

For students who are struggling to determine whether this relationship is proportional, suggest they create a table of values or graph based on the information given. An example table is shown.

- The Koreshan State Park in Estero, FL charges the following admission fees for vehicles entering the park.
 - \$4 for vehicles occupied by 1 person
 - \$5 for vehicles occupied 2 – 8 people
 - \$2 additional per passenger over 8 people

- Identify pairs of values that are comparing the cost of admission into the park for a vehicle to the number of people in the vehicle.

Number of People in the Vehicle	Cost in Dollars
1	4
2	5
3	5
4	5
5	5
6	5
7	5
8	5
9	7
10	9

- Determine whether the relationship between the two quantities is proportional. If it is proportional, state the constant of proportionality. If it is not proportional, explain why not.

The relationship between the number of people in the vehicle and the cost is not proportional because the ratio of y to x is not equivalent for all pairs of values.

3. Aluminum cans can be recycled instead of being thrown in the garbage. Some states pay people for returning cans so the aluminum can be reused for other products. The weight of 10 aluminum cans is 0.16 kilograms (kg). The weight of 25 aluminum cans is 0.4 kg.

- a. Identify pairs of values of the two quantities being compared.

Number of Cans	Weight of Cans (kg)
10	0.16
25	0.4

- b. Determine whether the relationship between the two quantities is proportional. If it is proportional, state the constant of proportionality. If it is not proportional, explain why not.

The relationship between the number of cans and the weight of the cans is proportional. The ratio of y to x is equivalent for all pairs of values. The constant of proportionality is 0.016 kg per can.

11.5.4 TRY IT: DETERMINING IF A RELATIONSHIP IS PROPORTIONAL (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider sharing the cognate of proportional (proporcional) with Spanish-speaking students. Also consider creating a word wall of frequently used mathematical vocabulary for all students to refer to throughout the unit.



11.5.5 Wrap-Up: Ticket Out the Door

1. The table shows how loud a sound is, in decibels (dB), depending on how far away the listener is from it, in feet (ft).

Distance to Listener (ft)	Sound Level (dB)
5	85
10	79
20	73
40	67

Use the table to complete the statements.

The table ☐ does ☒ does not represent a proportional relationship.

- ☒ It does not have a constant of proportionality.
☐ The constant of proportionality is _____.

11.5.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Use students' responses to gauge their understanding of proportional relationships and a constant of proportionality. Apply the information gained from this assessment to correct misconceptions and determine future lessons.